

Steady State Cost in Competing Aggregation-Fragmentation Dynamics

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ABSTRACT

Aggregation and fragmentation among polymers has long been an area of active research and utility in physical chemistry. Drawing inspiration from the dynamics of social formulation, we make a preliminary effort at optimization among various potential fragmentation kernels to regulate the probability of gelation. This work has relevance for any work formalizing the formulation, dissolving, and splitting of groups of abstract individuals. Social groups are of particular interest, from musical subcultures to the regulation of corporate consolidation.

1 Introduction

Things fall apart, Keats wrote.¹ Of course, they also come together. How is a balance struck between the two? What does does such a balance cost?

Whether *homo economicus* or *homo sociologicus*, the coordinating and organizational capabilities of the human race are broadly agreed upon as one of our key defining features. W

We attempt

The Introduction section can use referenced texts, for example which calls references listed as bibitems in the `sample.bib` bibliography file. The introduction should expand on the background of the work (some overlap with the Abstract is acceptable) and explain your interests in the problem. The introduction should not include subheadings.

Aggregation is generally the transition

$$A_i + A_j \xrightarrow{K_{ij}} A_{i+j}$$

where: - A is a cluster - i and j are masses - K_{ij} is the rate of aggregation between clusters of mass i and mass j into a clusters of mass $i + j$ - K , then, is an infinite symmetric matrix $K_{i,j} = K_{j,i}$

The master equations for which are:

$$\frac{dc_k}{dt} = \frac{1}{2} \sum_{i+j=k} K_{ij} c_i c_j - c_k \sum_{i \geq 1} K_{ik} c_i \quad (1)$$

$$= \frac{1}{2} \sum_{i,j \leq 1} K_{ij} c_i c_j [\delta_{i+j,k} - \delta_{i,k} - \delta_{j,k}] \quad (2)$$

where δ is the Kronecker delta.

We propose additional ****disaggregation**** based on the general idea

$$A_k \xrightarrow{L_{ki}} A_i + A_{k-i}$$

where L_{ki} is the rate of disaggregation from clusters of mass k to clusters of mass $i, k - i$ such that L is ****not**** a symmetric matrix, as L_{31} is potentially valued and L_{13} is certainly 0. And with master equations:

$$\frac{dc_k}{dt} = \sum_{i \geq 1} [L_{ik} c_i (1 + \delta_{i,2k}) - L_{ki} c_i]$$

where the Kronecker delta allows us to double count when the cluster splits into two equal clusters of size k .

2 Methods: A model of something

Up to three levels of **subheadings** are permitted. Subheadings should not be numbered (this is done by using an asterisk after any subsection or subsubsection command).

The problem

The first section should be a clear definition of the problem statement. That is, what is the system to be modelled and (if applicable) what specific research questions are you after.

This is example of more text under a subsection. Bulleted lists may be used where appropriate, e.g.

- First item
- Second item

The model

A clear model, specifying the modeling approaches, assumptions, parameters, objectives (what it should be used for or not). Consider using a model card once we revisit the concept in class.

Third-level section

Topical subheadings are allowed so feel free to change the subsection or subsubsection titles. This is an example of a subsubsection.

3 Results

The results can contain subheadings as desired and can discuss the results directly.



Figure 1. Legend (350 words max). Example legend text for an example figure. Figures should be in pdf format unless it is a photo or archival image not in vector graphics, as is the case here (in which case jpg and png are acceptable).

4 Discussion

The Discussion should be succinct. It typically involve any big picture discussions not covered in Sec. 3, i.e., the Results section. This was an example of a section reference using the `label` and `ref` commands. Figures and tables can also be referenced in LaTeX using the `label` and `ref` commands, e.g. Fig. 1 and Tab. 2.

This discussion can therefore mention scientific consequences of the results, ideas for future work, stress approximations or simplifications that limit the applicability of the model, or discuss things you tried that did not work. The Discussion section must not contain subheadings.

Acknowledgements

Thanks to Laurent Herbert-Dufresne, VSCI in general, and the authors of *A Kinetic View of Statistical Physics*²

Additional information

Data availability statement: No real world data is used in this project

Code availability statement: Code is available at github.com.

References

1. Yeats, W. *The Second Coming*.
2. Krapivsky, P. L., Redner, S. & Ben-Naim, E. *A Kinetic View of Statistical Physics* (Cambridge University Press, 2010), 1 edn.

Table 1. Model Card

1. Model Details

The problem
Modeling approach
Degrees of freedom
Number of parameters
Implementation language
Relevant publications
License
Contact information

2. Intended Use

Primary intended uses
If forecasting, validation metrics
If applicable, data incorporated
Primary users
Out-of-scope uses

3. Key assumptions

Population size
Contact structure
Time structure and memory effects
Other

4. Metrics

Computational complexity
Variation approaches

8. Ethical Considerations

Potential risks and harms
Bias and fairness concerns
Mitigation strategies
Impact on stakeholders

9. Suggested next steps

Issues with implementations
Recommendations for use
Future work

Condition	n	p
A	5	0.1
B	10	0.01

Table 2. Legend (350 words max). Example legend text for an example table.